INNOVATION SPIKE 2026: Cross-Functional AI Hackathon

1. Summary & Philosophy

The goal of this event is to break down domain-specific silos (such as deep banking or complex pharma regulations) and focus instead on solving **cross-cutting, foundational technical bottlenecks** that apply across all of our enterprise projects.

Core Philosophy: Problem Over Solution

Rather than prescribing a rigid technical path, we want to define clear **business pain points**. Teams are strongly encouraged to bring their own creativity, select their preferred tools, and engineer unique solutions.

The topics are meant to be broad problem areas, and that the mentioned architectures or tech stacks are examples rather than strict requirements. Teams should pick a focused use case within a topic, explain why it has business value, and then scope a realistic MVP for the hackathon.

Technology, and other rules

- **The "Bring Your Own Stack" Rule:** No framework is mandatory. Teams may use any tool they see fit. However, **every team must explicitly justify their architectural choices** during the final presentation. This justification will form a key component of their technical evaluation.
- **Enterprise-Grade Focus:** We are not looking for simple, brittle wrapper scripts. The resulting solutions should display deep structural thinking around production-readiness, security, and scalability.
- Teams are very encouraged to develop an MCP Application **also** of whatever solution they develop. This time teams are bigger than previous hackathon, this can be handle by one or two people of the team, using Nitrostudio.
- Cursor in Automode only. We will evaluate that you develop good and efficiently.
- Every team will present their project and solution in 15 (timed and uninterrupted) minutes. This presentation should be think and prepare along the way with the development of the solution. Not needed to be a formal ppt, but should showcase and present successfully the work done. As the teams will be bigger than last hackathon, this work can be assigned to 1 person of the team.
- There will be 5 minutes for questions of the audience and then next team will proceed.

2. Universal Evaluation Criteria

Each project will be evaluated out of **100 points** based on the following five enterprise pillars:

Category	Weight	Description
1. Working Prototype & Functional Completeness	**25%**	Does the system actually run during the live demo? Are key edge cases handled? How robustly does it execute the end-to-end user workflow?
2. Problem Fit & Business Value	**20%**	How effectively does the solution address the designated business paint point? Is there a clear, articulable path to driving real enterprise value?
3. Technical Design & Architecture	**20%**	Is the codebase modular, clean, and easily maintainable? Are technical choices (frameworks, storage layers, libraries) logical and well-justified?
4. AI Quality, Accuracy & Grounding	**15%**	Does the system minimize hallucinations and show reliable accuracy? How solid is the retrieval quality, semantic mapping, and prompt engineering?
5. Security & Responsible Data Handling	**10%**	How safely does the system handle sensitive information (PII/PHI)? Are encryption methods, vault isolation architectures, and access privileges soundly constructed?
6. UX, Demo Quality & Innovation	**10%**	Is the user interface seamless, responsive, and intuitive? Does the live presentation deliver a polished, creative "wow" factor?

3. The 5 Hackathon Challenge Topics

Topic 1: Anonymization - Privacy Preserving Data Access

Description & Objective

The objective of this challenge is to design and build a high-performance **Reversible Pseudonymization Pipeline**. The pipeline must ingest datasets containing sensitive PII/PHI (names, phone numbers, Aadhaar IDs, financial records), identify and replace these identifiers with **format-preserving, deterministic tokens**, and output the masked dataset into a queryable big data format.

Simultaneously, the pipeline must secure **Privacy Preserving Data Access** . Authorized users must be able to securely query the database or reverse the tokens, while downstream analytical tools and LLMs interact *only* with the safe, tokenized data.

Input & Output Specifications

- **Inputs:** Mixed unstructured/semi-structured files (CSVs, unstructured medical transcripts, bank statements) featuring raw names, phone numbers, Indian Aadhaar numbers, age, diseases, and medication lists.
- **Outputs:** * **Anonymized Layer:** Sanitized, highly compressed Apache Parquet or Delta Lake format, ready for analytics.
    - **The Vault Storage:** An isolated, encrypted mapping database (e.g., AES-SIV or salted hashes with symmetric key storage) facilitating secure token-to-raw value translation.
    - **Conversational Interaction:** An active AI intervention terminal where the system prompts the administrator when it encounters ambiguous boundaries (e.g., *"The term 'Dr. Amol' is identified as a custom entity. Should this be tokenized under PHI rules?"*).

Pros, Cons, and Feasibility Analysis

- **Expert Opinion:** Direct your teams to focus heavily on **Linkability & Referencing**. Ensure they prove that they can run a complex, multi-table query (e.g., *"Compare average healing times for clinical trials of Drug X across age buckets 30-50"*) directly on the *pseudonymized* dataset, and that the query yields the exact same logical result as if it were run on the raw, sensitive database.

Topic 2: Turning Messy Enterprise Documents Into Usable Data

Description & Objective

Enterprises process enormous quantities of administrative documentation locked in incompatible formats: multi-column PDF medical papers, tabular financial Excel sheets, raw txt files, and handwritten logs.

To solve this, teams must build a robust **General-Purpose Multimodal Ingestion Engine** that structures diverse files into a single, uniform, easily usable place.

Key Ingestion Scope & Guidelines

To prevent scope creep in a tight 2-day hackathon, **this track is strictly limited to Text files, multi-column PDFs, complex Excel spreadsheets, and raw Images of handwritten notes**. Video and audio parsing are excluded from this challenge and belong under VoiceAI (Topic 3).

A **SUGGESTION** of the ingestion engine can be a classic big data **Medallion Architecture**:

1. **Bronze Layer (Raw Archive):** Stores original ingested documents as raw blobs.
2. **Silver Layer (Structured Schemas):** Parses files and runs them through a rigid schema validation layer (e.g., Pydantic or JSON schema validation). This converts nested tables, hand-written text, and raw paragraphs into highly validated JSON records, logging any structure violations.
3. **Gold Layer (Curated Analytics):** Converts validated JSONs into highly compressed, partition-based .parquet files organized logically by domain or date, optimized for blazing-fast analytical queries.

Input & Output Specifications

- **Inputs:** A messy batch of documents containing:
  - Invoices or medical receipts (Handwritten images/scans).
  - Financial ledgers (Complex multi-tab, nested Excel spreadsheets).
  - Academic or clinical records (Multi-column, complex layouts in PDF).
- **Outputs:**
  - Structured database rows stored in organized Parquet partition directories (e.g., .../domain=healthcare/year=2026/month=06/*.parquet).
  - A real-time **Pipeline Health Dashboard** displaying total file sizes ingested, layout extraction accuracy, validation failure rates, and processing latency.

Pros, Cons, and Feasibility Analysis

- **Expert Opinion:** Advise teams to avoid writing custom regex or layout rules for every document type. Instead, they should build a **unified multimodal parsing prompt** leveraging Gemini 2.5 Flash, which converts layout metadata directly into a strict, validated JSON structure. Focus grading heavily on how gracefully the pipeline handles schema mismatches or corrupted files.

Topic 3: VoiceAI-Driven Agentic Workflow Orchestration

Description & Objective

Develop an active, highly communicative voice-agent capable of orchestrating backend system workflows using natural spoken language. The agent should sound human-like, have low latency, and be capable of initiating actions like correcting database records, filling forms, booking appointments, and sending email summaries based entirely on verbal inputs.

[Human Voice Input] ---> [STT] ---> [Agentic Router (Tool Calls)] ---> [DB/Email API / TTS Output]

Input & Output Specifications

- **Inputs:** Real-time or recorded voice (audio stream).
- **Outputs:**
  - Triggered state changes (e.g., API calls to database, emails sent via Mock SMTP).
  - Real-time voice feedback with natural cadence, low latency (under 1.5 seconds), and a human-like tone.
  - Visual representation of the "agentic thought process" on a live monitor during the conversation.

Pros, Cons, and Feasibility Analysis

- **Expert Opinion:** Teams should prioritize **"Tool Calling" (Function Calling)**. Rather than just transcribing voice to text, the agent must extract entities (e.g., dates, phone numbers, policy IDs) and invoke specific functions to change state.

Topic 4: Intelligent Semantic Retrieval & Interactive Analytics Dashboard

Description & Objective

Once data is digitized (from Topic 2) or anonymized (from Topic 1), users need an intuitive interface to extract actionable insights. Teams must build an advanced semantic search and visualization dashboard (RAG) that allows non-technical domain experts (e.g., clinical doctors or credit risk officers) to query complex datasets in natural language, view interactive analytical charts, and conduct comparative multi-turn analyses.

[NL Query: "Compare patient recovery trends..."] ---> [RAG / Vector Database] ---> [Structured Insights + Chart Generation]

Input & Output Specifications

- **Inputs:** User-supplied natural language questions (e.g., *"Show me a comparison of patients aged 30-50 who took Medicine A vs Medicine B"*).
- **Outputs:**
  - Semantic citations pointing to the exact source documents (addressing hallucination worries).
  - Dynamic, interactive data visualizations.
  - A clean, responsive dashboard interface containing logical drill-down paths.

Pros, Cons, and Feasibility Analysis

- **Expert Opinion:** The winning differentiator for this track will be **Dynamic Schema Mapping & Chart Generation**. Instruct teams not to hardcode visual outputs, but to use the LLM to generate structured configurations (like Vega-Lite or JSON schemas) that the UI dynamically renders into beautiful charts.

Topic 5: The Guardrail Plug: Deterministic Auditing & Hallucination Defense

Description & Objective

LLMs are notoriously prone to "hallucinations," making them risky for highly regulated sectors. Teams must build an independent **Auditing & Logging Middleware (a plug-and-play SDK)**. This tool should intercept inputs and outputs of *any* given AI agent, record step-by-step reasoning, trace which factual documents were referenced, and block/flag any answers that cannot be deterministically proven by the underlying knowledge base.

[Agent Output] ---> [Auditing Middleware (Source Check / Rule Engine)] ---> [Passed Verified Output] OR [Blocked/Flagged]

Input & Output Specifications

- **Inputs:** Raw inputs/outputs from an active LLM agent, accompanied by the vector database chunks that were retrieved.
- **Outputs:**
    - A mathematical "Factual Trust Score" (from 0.0 to 1.0) indicating the degree of grounding in facts.
    - A visual execution graph showing the exact chain-of-thought, tool invocations, and sources.
    - An automated fallback output (e.g., *"I cannot verify this information because..."*) if a hallucination threshold is breached.

Pros, Cons, and Feasibility Analysis

- **Expert Opinion:** To make this project stand out during judging, teams should build a **Real-Time Auditing Dashboard** that shows a side-by-side comparison of: Raw Agent Thought Processes vs. Fact-Audited Ground Truth Paths, highlighting exactly where a hallucination was caught and redirected.